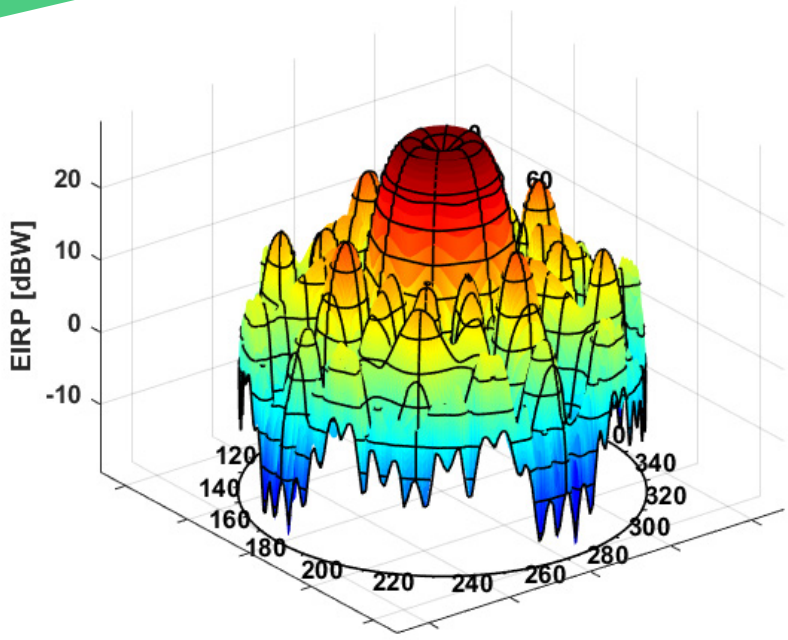
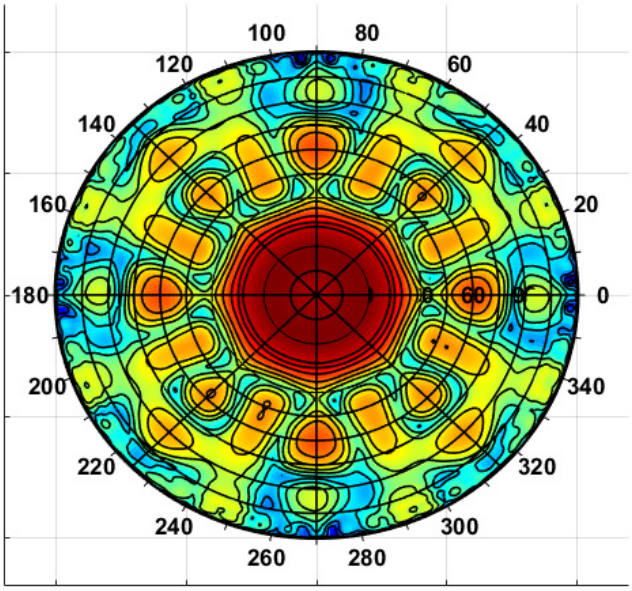
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Issue 1.0 | May 2024





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DOCUMENT CHANGE RECORD

REASON FOR CHANGE	ISSUE	REVISION	DATE
First version of the document	1	0	May 2024

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FOREWORD

This document has been prepared under the coordination of the Directorate-General for Defence Industry and Space of the European Commission (DG DEFIS). The model presented in the next pages has been developed by the Joint Research Centre (JRC) of the European Commission, within the Technologies for Space, Security and Connectivity Unit. The work leading to the publication of the document is the result of an intense collaboration between JRC and ESA/ESTEC, and has built upon previous work developed by ESA. ESA and EUSPA also contributed to the review of the document and of the metadata provided in attachment.

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1 INTRODUCTION

The Space Strategy for Europe, released in 2016[1], has as one of its main objectives to foster a globally competitive and innovative European Space Sector. A key element to accomplish that objective is the use and validation of Galileo in Space. Along this line, the Galileo Programme has formalized the definition of a Space Service Volume (SSV) as integral part of the Mission for the Galileo 2nd Generation.

The use of Galileo, as GNSS in general in Space, particularly for certain missions (MEO, GEO, HEO, Cis-Lunar and beyond), can be best supported by knowledge of the navigation antenna patterns, or a representative model thereof, including side lobes. Different studies on high orbit navigation ([1], [2], [3]) rely on the knowledge of the received signal power [5], and refer to the expected constellation Antenna Gain Pattern (AGP), with particular focus on the side lobes.

At International level, effort is ongoing on the definition of an Interoperable Global Navigation Satellite Systems Space Service Volume [15] under the International Committee on GNSS (ICG). In this respect, ICG explicitly recommended in 2019 the "*release of the GNSS transmit antenna patterns or equivalent representative modelling information, including the side lobes, by all GNSS service providers, in order to fully exploit the potential of GNSS for space users, including missions to the Moon and beyond*"[2].

Last but not least, European Industry and Academia have raised their interest on the knowledge on the characteristics of the Galileo Navigation antenna. The Programme is publishing this document to also satisfy such interest.

Within Figure 1, a representation of the typical space service volume geometry is provided, referring in particular to high altitude orbits (i.e. user altitude above MEO) and antenna nadir pointing. The right half of Figure 1 describes the dynamic evolution of a Galileo satellite rising above the Earth, as seen from a user on GEO orbit, at four specific anomaly angles θ_i, with $i = \{1,2,3,4\}$. Those correspond to a set of off-boresight angles α_i, with $i = \{1,2,3,4\}$, as seen from the Galileo satellite antenna (whose boresight is aligned with nadir). Such angles identify different pattern regions observed by the user during the rising of the Galileo MEO satellites out of the Earth shadow. The figure on the bottom left side extends the example for the user altitude spanning from GEO to Moon. As the altitude increases, the maximum off-borasight angle is observed for smaller rising angles.

[1] COM(2016) 705 final, Space Strategy for Europe, 26.10.2016

[2] United Nations, Committee on the Peaceful Uses of Outer Space, Fourteenth meeting of the International Committee on Global Navigation Satellite Systems, 2019, A/AC.105/1217

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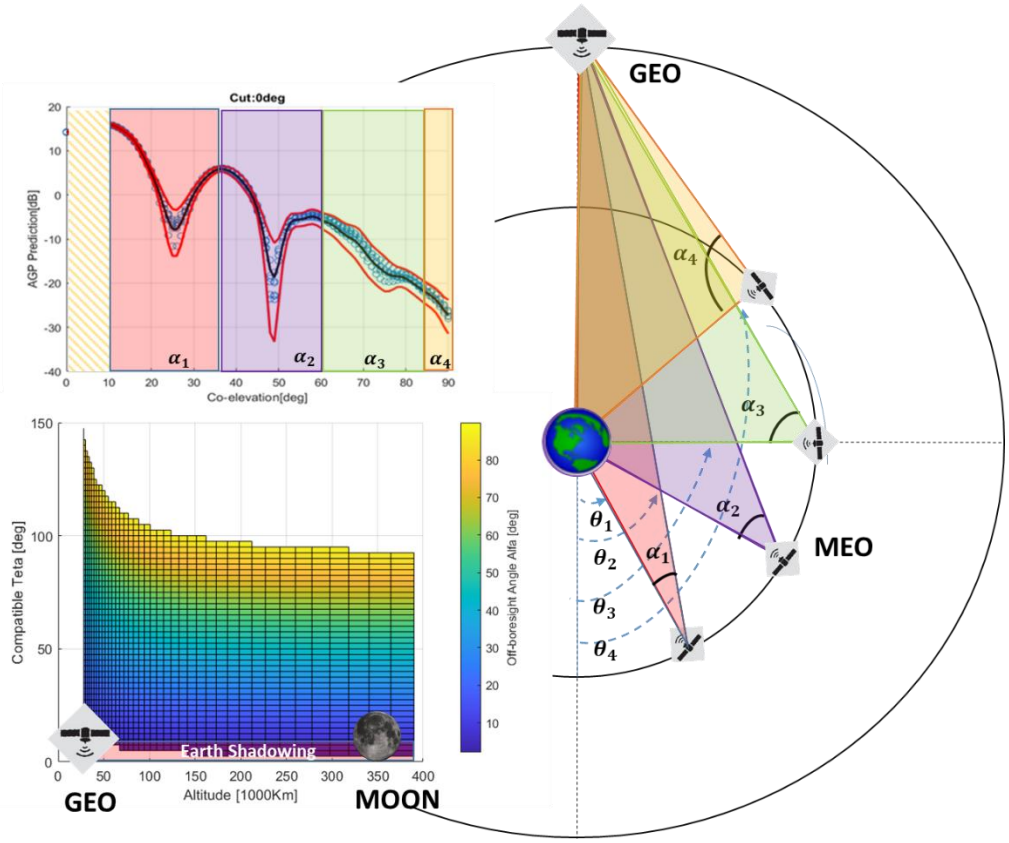


Figure 1. Space Service Volume geometry summary

This document presents the *Galileo Reference Antenna Pattern (GRAP) for* the Galileo first generation Full Operational Capability (FOC) satellites, as the model representing the Equivalent Isotropic Radiated Power (EIRP) that users can expect based on measurements and information established at system level before deploying the satellites. The obtained model has been cross-validated with available measurements of the integrated satellite, as explained later. The intention is to provide the user community with a representative model derived from the information retrieved at the initial stage from the measurement campaigns performed on all the Galileo FOC satellite[3] antennas. Actually, the exploited procedure, presented in details in [1], reconstructs ([7],[8],[14]) and then statistically aggregates ([11]) all information for the user into the *GRAP* mean, GRAP lower and upper bound, with the possibility to extend the performance to all possible realizations and observations compatible with the defined interval.

This document is the first version and was obtained exploiting the best knowledge available within the Galileo Programme at the time of publication. The model presented here was conceived to allow for successive updates when new data (i.e., from additional measurements) is available in future. In that case, the model metadata would be updated for their further exploitation by the Galileo space users.

This document is structured as follows:

- Chapter 2 provides the model background
- Chapter 3 provides the model description and the algorithm to retrieve the model from the metadata
- Chapter 4 introduces the model metadata

[3] Galileo FOC satellites are those referred to as GSAT02XX

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2 MODEL BACKGROUND

The *Galileo Reference Antenna Pattern (GRAP)* model is the result of the processing of the measurements dataset collected through the measurement campaigns performed on all the antennas before their integration on the Galileo FOC satellites. The obtained model has then been cross-validated up to 53 degrees of co-elevation with the available measurements of the antenna after the integration.

A 2D antenna radiation pattern reconstruction procedure, derived from antenna characterization techniques ([6], [12], [13]) has been considered to mitigate unexpected discontinuities and reduce measurement degradation effects. As a first step, a Power Spectral Density (PSD) weighting of the Continuous Wave (CW) antenna measurements was applied in order to resemble the power distribution of the modulated signal in the frequency band. Then, the spatial reconstruction was performed on the basis of a spherical harmonic expansion in order to obtain a smooth representation coherent with the expected physical behaviour.

The Galileo FOC Satellite antenna radiation patterns are processed as observations nested or clustered within different subjects at the bottom level, and belonging to the same constellation population at the top level [11]. This approach, the details of which are introduced in [1], aims at extending the model to a more general representation, which statistically takes into account possible antenna realizations and residual errors through a correspondent variance estimation.

The link budget resulting into the EIRP (see next section) in the Space Service Volume ([9], [10]) can take into account the expected deviation from the nominal value at time of the characterization. Moreover, it is possible to consider the model as an initial a-priori distribution (first and second moments) that can be integrated with other data potentially available to the users.

As it will be detailed in the next chapter, the model covers the various Galileo frequency bands (E1, E5a, E5b and E6.)

In the next Chapter 3 the definition of the model format and the corresponding upper and lower bounds are provided, while the corresponding model metadata are introduced in Chapter 4 and can be found in attachment to the document.

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3 MODEL DESCRIPTION

The Galileo Reference Antenna Pattern is provided in terms of Equivalent Isotropic Radiated Power (EIRP) with respect to the azimuth and co-elevation angles. The EIRP as provided in this document does directly correspond and ensure that the minimum user received power at ground level does correspond to the Galileo commitments for E5ab-IQ, E6-BC and E1-BC signals as established in [AD.1].

The EIRP is defined for each azimuth and elevation angle as follows:

$$\boldsymbol{EIRP_{GRAP,f,dBW}}(\theta,\varphi) = Pt_{f,dBW} + \boldsymbol{G_{f,dB}}(\theta,\varphi)$$

Where $Pt_{f,dBW}$ is the transmitted power for the specific signal and frequency band *f* (in dBW) and $\boldsymbol{G_{f,dB}}(\theta,\varphi)$ is the antenna gain expressed as function of co-elevation θ and azimuth φ. The EIRP model confidence intervals (*CI*) can be computed as follows:

$$\boldsymbol{EIRP_{CI,f,dBW}}(\theta,\varphi) = \boldsymbol{EIRP_{GRAP,f,dBW}}(\theta,\varphi) \pm 2\boldsymbol{\sigma_{GRAP,f,dB}}(\theta,\varphi) \quad (1)$$

Where $\pm 2\boldsymbol{\sigma_{GRAP,f,dB}}(\theta,\varphi)$ are the upper and lower bounds (95%) of the EIRP, respectively.

A 3D graphic representation of the obtained reference pattern for E1-BC, $\boldsymbol{EIRP_{GRAP,f,dBW}}(\theta,\varphi)$, is provided in the following figure together with the correspondent polar view.

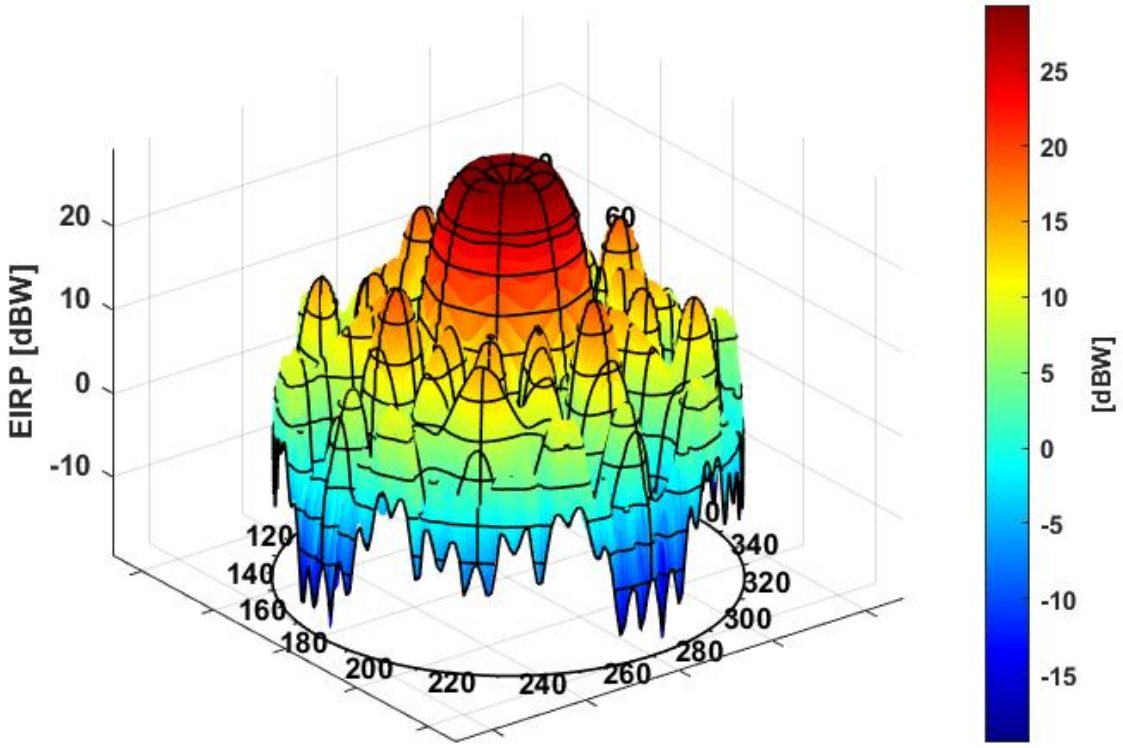


Figure 2. Galileo E1-BC 3D Constellation [dBW] $\boldsymbol{EIRP_{GRAP,E1-BC,dBW}}$

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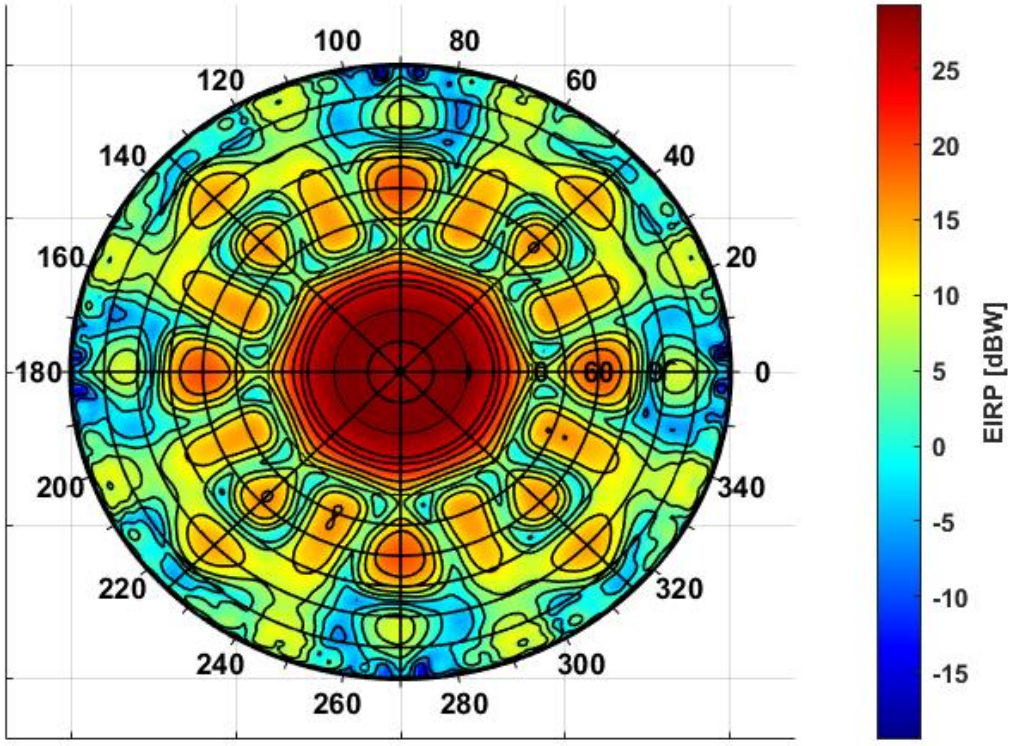


Figure 3. Galileo E1-BC 3D Constellation [dBW] $EIRP_{GRAP,E1-BC,dBW}$ - polar view

The polar plot of the corresponding 95% pattern variation $\boldsymbol{EIRP}_{CI,f,dBW}(\theta,\varphi)$ is provided in the next figure.

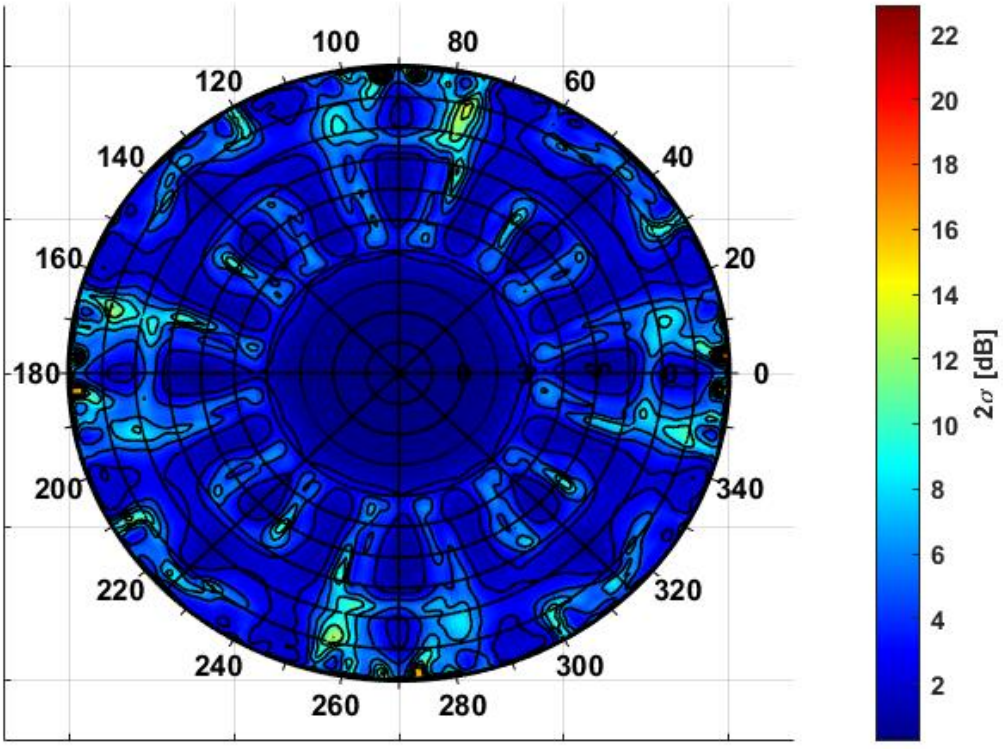


Figure 4. Galileo E1-BC 3D Constellation EIRP expected 95% variation [dB] $2\sigma_{\mathrm{GRAP,E1-BC,dB}}(\boldsymbol{\theta},\boldsymbol{\varphi})$- polar view

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Some representative cuts (0, 90, 135 and 270 deg azimuth) are represented below in order to appreciate the capability of the lower $-2\boldsymbol{\sigma}_{\mathbf{GRAP,f,dB}}$ and upper $+2\boldsymbol{\sigma}_{\mathbf{GRAP,f,dB}}$ bounds to envelop the expected EIRP excursion around the reconstructed constellation pattern.

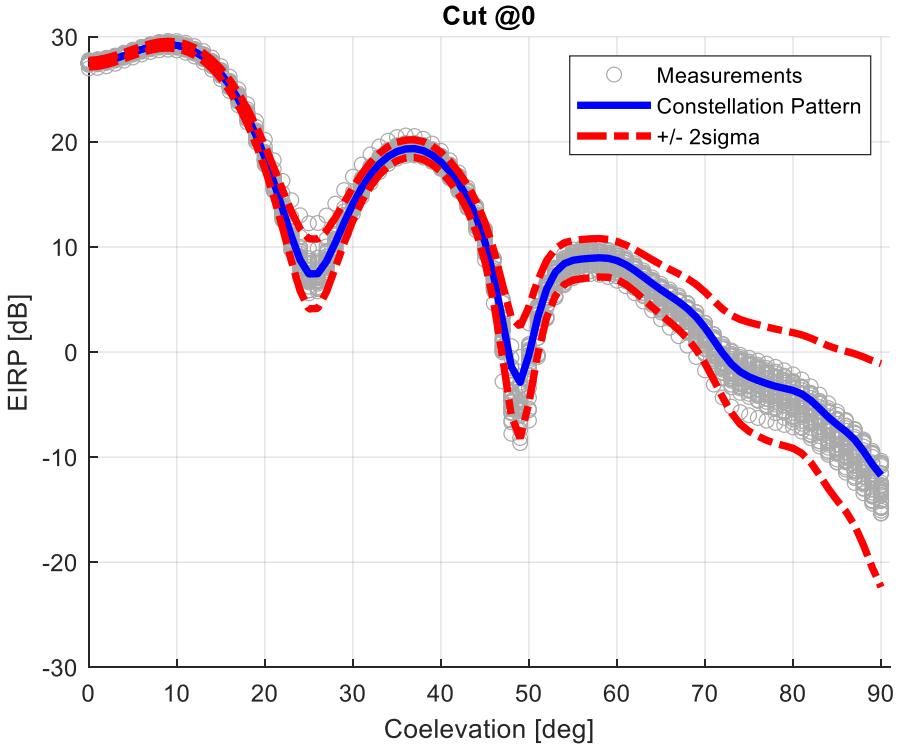


Figure 5. Galileo E1-BC $EIRP_{GRAP,E1-BC,dBW}$ (cut @0 deg azimuth) with expected 95% variation [dB]

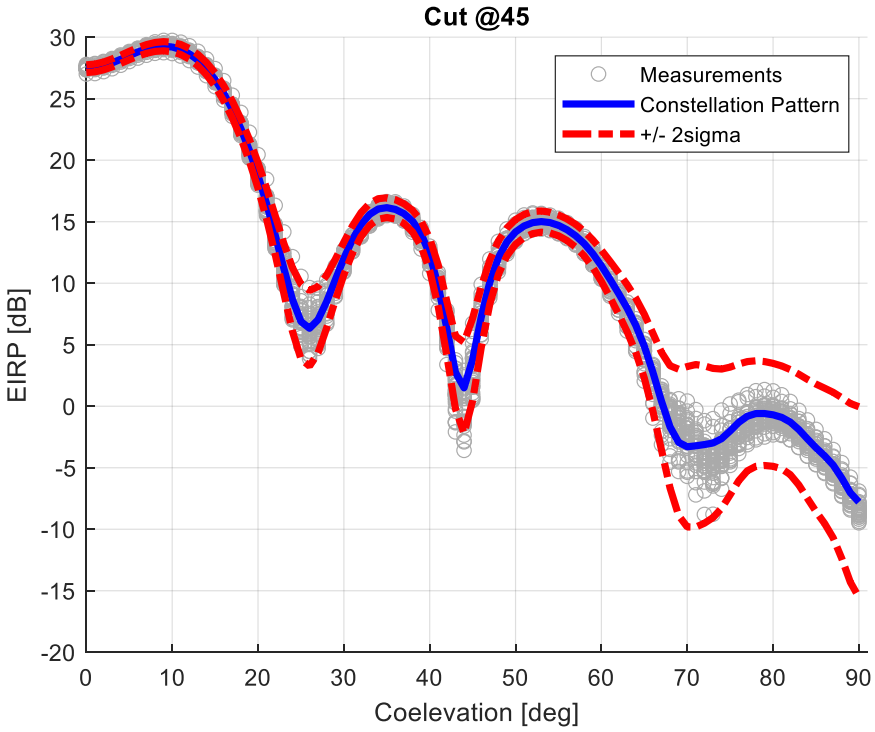


Figure 6. Galileo E1-BC $EIRP_{GRAP,E1-BC,dBW}$ (cut @45 deg azimuth) with expected 95% variation [dB]

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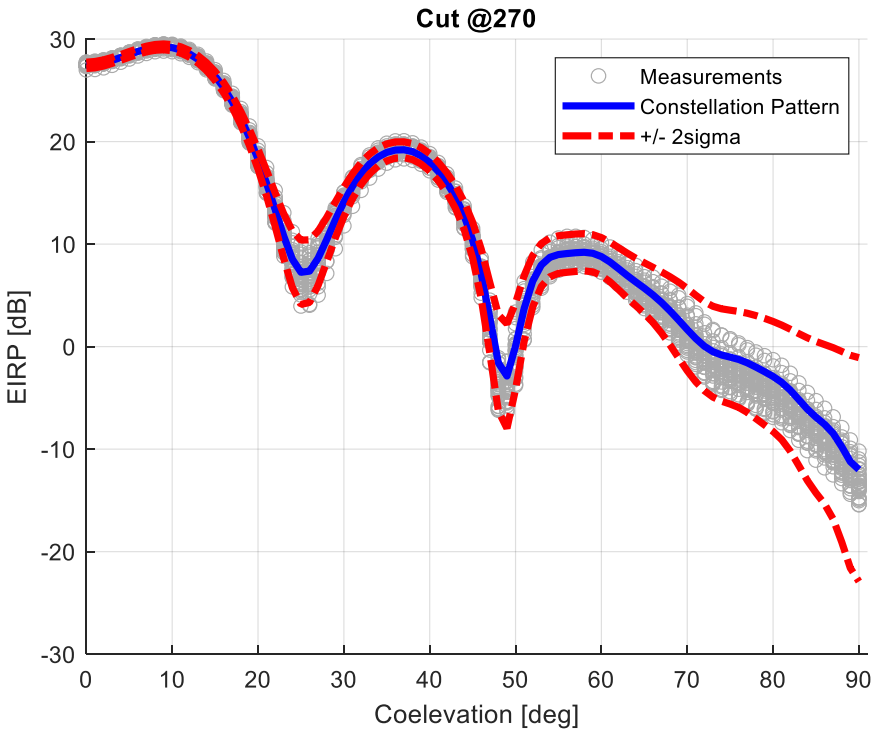


Figure 7. Galileo E1-BC $EIRP_{GRAP,E1-BC,dBW}$ (cut @270 deg azimuth) with expected 95% variation [dB]

The EIRP distribution shown in the figures above and made available to the user through metadata defined in the next section is coherent with the received power levels on ground for the E1-BC signal as per [AD1]. The same applies for E5a (E5a-IQ signal), E5b (E5b-IQ signal) and E6 (E6-BC signal).

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4 MODEL METADATA

The different elements presented in Chapter 3 are provided to the users as metadata in the files annexed to this document and detailed in the tables below. The files, provided in .xls format and organized as described in the table below, can be accessed/saved from the attachments panel of the pdf file reader.

Symbol	Description	Size	Ref. File
$\boldsymbol{EIRP}_{\boldsymbol{GRAP,f,dBW}}(\theta,\varphi)$	EIRP [dBW]	[91x361], 1deg	GRAP_File_****.GRAP_EIRP_dBW_****.xls
$\boldsymbol{EIRP}_{\boldsymbol{CI,f,dBW}}(\theta,\varphi)\big\vert_{(+)}$	EIRP Upper Bound (95%) [dBW]	[91x361], 1deg	GRAP_File_****.GRAP_UB_dBW_****.xls
$\boldsymbol{EIRP}_{\boldsymbol{CI,f,dBW}}(\theta,\varphi)\big\vert_{(-)}$	EIRP Lower Bound (95%) [dBW]	[91x361], 1deg	GRAP_File_****.GRAP_LB_dBW_****.xls

Table 1: Model Metadata – GRAP and variances in dB

where ****. is replaced by {E1__, E5a_, E5b_, E6__} according to the target frequency band.

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5 APPLICABLE DOCUMENTS

[AD1] European Union, Galileo Open Service Signal in Space Interface Control Document, Issue 2.0, January 2021

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6 LIST OF REFERENCES

[1] Menzione, F., Sgammini M., Paonni M., Reconstruction of Galileo Constellation Antenna Pattern for Space Service Volume Applications, JRC135110, 2023

[2] Enderle, W., Gini, F., Schönemann, E., Mayer, V., PROBA-3 Precise Orbit Determination based on GNSS observations, Conference: 32nd International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2019)

[3] Marquis, W. and Reigh, D., The GPS Block IIR and IIR-M Broadcast L-Band Antenna Panel: Its Pattern and Performance, NAVIGATION, vol. 62, n. 4, 2015

[4] Moreau, M., et. Al., Results from the GPS Flight Experiment on the High Earth Orbit AMSAT AO-40 Spacecraft, Proceedings of the ION GPS 2002 Conference, Portland, OR, 2002.

[5] GPS Space Service Volume Ensuring Consistent Utility Across GPS Design Builds for Space Users, NASA May 27, 2015

[6] Migliore, M. D., Soldovieri, F., and Pierri, R., Far-field antenna pattern estimation from near-field data using a low-cost amplitude-only measurement setup, IEEE Trans. Instrum. Meas., vol. 49, no. 1, pp. 71-76, Feb. 2005

[7] Allende-Alba, G., Thoelert, S., Reconstructing antenna gain patterns of Galileo satellites for signal power monitoring, GPS Solutions 24, 22 (2020)

[8] Allende Alba, G., Thoelert, S., An analysis of the on orbit performance of Galileo satellite antennas using reconstructed gain patterns, GPS Solutions (2020)

[9] Shehaj, E., Capuano, V., Botteron, C., Blunt, P. and Farine, P.-A., GPS Based Navigation Performance Analysis within and beyond the Space Service Volume for Different Transmitters' Antenna Patterns, in 13th PEGASUS-AIAA Student Conference, Berlin, 2017

[10] Ji, G.-H.; Kwon, K.-H.; Won, J.-H., GNSS Signal Availability Analysis in SSV for Geostationary Satellites Utilizing Multi-GNSS with First Side Lobe Signal over the Korean Region, Remote Sens. 2021

[11] Fitzmaurice, G., Davidian, M., Verbeke, G., Molenberghs, G., Longitudinal Data Analysis, 2008

[12] Zheng, J., Chen, X., Liu, X., Zhang, M., Liu, B., and Huang, Y., An Improved Method for Reconstructing Antenna Radiation Pattern in a Loaded Reverberation Chamber, in IEEE Transactions on Instrumentation and Measurement, vol. 71, pp. 1-12, 2022

[13] Reza Behjoo, H., Pirhadi, A., Asvadi, R., Efficient spherical near-field antenna measurement using CS method with sparsity estimation, IET Microwaves, Antennas and Propagation, Vol 13, Issue 11, Sept. 2019

[14] Xu, Q., Huang, Y., Xing, L., Song, C., 3D Antenna Radiation Pattern Reconstruction in a Reverberation Chamber Using Spherical Wave Decomposition, IEEE Transactions on Antennas and Propagation, Volume: 65, Issue: 4, April 2017

[15] International Committee on GNSS (ICG), The Interoperable Global Navigation Satellite Systems Space Service Volume, Second Edition, 2021, ST/SPACE/75/Rev.1

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